

**Measuring NHL Organizational Player Development:
An Organizational Development Index (ODI) for Drafted Prospects
(2010–2020)**

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Table of Contents

Abstract.....	3
1. Introduction.....	4
2. Literature Review.....	5
3. Methodology.....	7
3.1 Data Collection.....	7
3.2 Evaluation Window.....	8
3.3 Organizational Credit.....	9
3.4 Player Value Metric.....	10
3.5 Expected Pick Value.....	11
3.6 The Organizational Development Index (ODI).....	13
4. Results.....	14
5. Discussion.....	19
6. Limitations.....	21
7. Future Research.....	25
8. Conclusion.....	27
Appendices.....	28
Annotated Bibliography.....	30

Abstract

I am a New York Islanders fan, and that is genuinely where this project started. The organization's history with developing players is inconsistent: Rick DiPietro is the example everyone points to as a failure, but Mathew Barzal came out of that same system and became a legitimately elite player. I wanted to know whether that inconsistency was real, and how it actually compared across the rest of the league, rather than relying on reputation and memory. It became clear this was not just a question for a fan to be curious about. An organization could look at a tool like this and identify, concretely, where its development system is underperforming rather than guessing.

I built the Organizational Development Index (ODI) to answer that question directly. The framework measures how efficiently an NHL organization converts a drafted player into real NHL value, relative to what that player's draft position and role should be expected to produce. The dataset covers every NHL Entry Draft selection from 2010 through 2020, over 2,300 players, collected through a web scraper I built myself and evaluated using Hockey-Reference's Point Shares metric within fixed, position-adjusted evaluation windows.

The most important decision in this project was refusing to let a team keep permanent credit for a player it no longer develops. I built the model to calculate value two ways: Development Credit, which credits the drafting team regardless of trades, and In-Organization Credit, which only counts value produced while the player was actually still on that roster. This was not a theoretical exercise. For example, Nashville's ranking collapses once Seth Jones and Samuel Girard's post-trade production is removed. Washington's case is more direct still: Filip Forsberg was traded before he played a single game for them, so any credit Washington receives for him primarily reflects talent identification rather than organizational development.

I did not simply trust my own first result. I went back through the data and found two real problems before drawing any conclusions: a scraping error that had zeroed out production for thirteen players, including established NHL player Nicolas Roy, and a separate error that had silently recorded zero value for every goalie in the entire dataset. I found both, corrected both, and re-ran the analysis to confirm the fix held.

What this project shows is that aspects of organizational development can be measured, not just assumed from reputation. Across two independent evaluation windows and a set of tests designed specifically to try to break the result, Anaheim, Boston, and Carolina consistently convert their draft picks into more value than expected. I am not claiming this model accounts for everything. Injuries, draft-class strength each year, and the number of opportunities a player is given all matter, and I address those limitations directly rather than pretending they do not exist. The core finding, however, reflects what the data actually shows, not a guess.

1. Introduction

I have noticed a consistent problem in how people talk about which organizations are good at developing players. Fans and analysts will point to specific names but rarely step back and look at the pattern across an entire organization's history. Edmonton drafted Connor McDavid first overall, and it is tempting to call that evidence the Oilers are great developers. It is not. A player selected first overall is already expected to become a superstar. Nothing about that outcome tells you anything about the organization's actual development ability, because the same standard cannot be applied to a first-round pick and a fourth or fifth-round pick. The expectations are completely different, and treating every good outcome as equal evidence of development skill misses that entirely.

The same problem shows up from the other direction. Analysts will often reason backward: a team is playing well right now, so the players it drafted or acquired must have been developed well. That is not necessarily true either. With as many players drafted every year as there are, and as much roster turnover as exists across a league, focusing only on the biggest names and the biggest failures is not a complete or reliable way to evaluate an organization. When people say a team like Carolina is good at development, that claim deserves to be backed by something more than the fact that they are winning this season. It should be backed by what they have done, consistently, over a period of years. The reverse is true as well. Detroit has not developed or drafted well for most of the past decade, and that pattern shows clearly in the data this project collected, not just in reputation.

People talk about which NHL teams are good at developing players the same way they talk about anything else in the sport: with a lot of confidence, and not much behind it. Carolina's good at this. Detroit isn't. Everyone seems to agree, but nobody's actually checking. That's the part that bothered me. Not another opinion added to the pile, an actual answer. So this is the question this project was built to answer: how efficiently does an NHL organization convert the players it drafts into NHL value relative to what their draft position predicts, and how much of that value the organization actually keeps versus loses through trades. I wanted something built on statistics and collected data, not on the same handful of names people keep repeating because they sound right.

2. Literature Review

Before building this framework, I looked closely at what already exists in this space, specifically to figure out what each tool actually answers and where the real gap was.

Draft pick value charts, most notably the curve built by analyst Eric Tulsky, tell you how a player selected at a given position in the draft is expected to perform on average. I referenced Tulsky's

curve directly while working out the reasoning behind this project's own evaluation windows, since it deals with the same underlying question of when a drafted player starts producing real value. But a pick value chart stops there. It does not ask whether a player drafted at that position by this specific team performed to that expectation, and it does not track whether that team has done so consistently over time. It tells you what should happen at a given draft slot in general. It says nothing about whether a specific organization is actually delivering on that expectation.

WAR, and similar all-in-one value metrics, measure something different entirely. They target individual player value, not organizational development ability. A high WAR tells you a player performed well, but it credits that performance to a single team for his entire career and does not track how his value shifts across organizations if he is traded. This project does not fully separate an organization's scouting judgment from its development work either; that distinction remains genuinely difficult for any model in this space, and is addressed directly in the Limitations section. What this project adds, that WAR does not, is a way to track a player's value specifically to whichever organization actually controlled him at the time it was produced, rather than crediting his entire career to the team that drafted him regardless of what happened afterward. That is a narrower, more tractable question than WAR is built to answer, especially once a player changes teams.

Prospect rankings ask a third, still different question: how did this player perform against his peers before he was ever drafted. That can tell a team a prospect has real tools and translatable talent. It cannot tell you how well that player was developed once he entered a professional organization, because it is measuring him against players his own age, not against NHL competition.

Taken together, none of these tools, individually or combined, answer the question this project is built around. Each one either evaluates the pick, the player, or the prospect in isolation, but none of them evaluates the organization itself, over time, against a real and specific expectation. That is the gap this project is built to fill.

3. Methodology

3.1 Data Collection

Manually looking up statistics for over 2,300 players across eleven draft classes was never realistic. Pulling every player's name from a draft class page is simple enough, but each player also needed real statistical detail behind them, not just a name, and doing that individually for more than two thousand players would have taken an unreasonable amount of time. Building a scraper was the only practical way to do this on the scale the project needed.

I chose Hockey-Reference specifically because its Point Shares metric is the most accurate value measure I found. It accounts for more than just points, goals, and assists, which was important given what this project was trying to measure. I verified the scraper's reliability directly rather than assuming it was correct: I checked that it was pulling complete statistical records for each player, not just partial data, and I cross-referenced per-year draft pick totals against an independent source, hockeydb.com, to confirm the scraper was not silently missing players.

I also built the scraper to be resumable from the start, since a single full run could take well over an hour, sometimes closer to two, and I had no interest in losing that progress to a crashed terminal or a dead laptop battery partway through. I initially tested running two scraping processes at once to save time, and separately tested a faster request interval than what I ultimately used. Both approaches caused the site to rate-limit and reject requests. I settled on a single process, running

sequentially, at a two-second interval between requests. It was slower than the alternatives I tried, but it was reliable, and getting complete, trustworthy data mattered more to me than saving an extra twenty or thirty minutes.

3.2 Evaluation Window

I used two separate evaluation windows, six years and eight years post-draft, rather than just one. A fixed window matters because it gives every drafted player, regardless of what year he was actually drafted in, the same amount of time to actually show what he became. A player drafted in 2010 is followed through the 2016 season under the six-year window, and a player drafted in 2020 is followed through the current 2026 season under that same six-year standard, so every draft class is judged on equal footing. The eight-year window follows the same logic, just with two more years of development time factored in, which matters for positions like defense and goaltending that tend to take longer to reach their NHL ceiling. Because the eight-year window requires eight full seasons of data, it only includes draft classes from 2010 through 2018, the 2019 and 2020 classes have not yet had enough time to complete that window and are excluded from it, while the six-year window includes all eleven classes, 2010 through 2020.

I also chose 2010 through 2020 specifically because it gave the project the broadest usable player pool I could reasonably work with, which mattered for sample size and for making sure the resulting numbers were actually statistically meaningful rather than based on too small a group of players. That said, this project is meant as more of a proof of concept, not a fixed, permanent range. The years themselves could be shifted or expanded in a future version, provided the additional draft classes are scraped and incorporated in the same way. Building it across 2010 through 2020 was

simply the most practical starting point, using data that already existed and was large enough to work with.

3.3 Organizational Credit

This distinction between the Development Credit and the In-Organization Credit ranking came out of a simple question I asked myself while looking at the first version of the results: what happens if a player gets traded? If a player is no longer on the team that drafted him, that team is no longer the one currently developing him. In hindsight it feels obvious, almost like something I should have accounted for from the start, but it genuinely was not obvious to me until I was already looking at real output and realized something was missing. Without that distinction, the model was only showing which organizations are good at recognizing talent at the moment of the draft. It was not answering how much of that talent's later value the organization actually retained, which is the second half of this project's research question.

Not every player stays with the team that drafted him for his entire career. Not everyone is Sidney Crosby. Players get traded, and that movement must be factored into how credit is assigned, or the model is only measuring scouting, not development.

To handle this, I built the model to calculate two separate versions of every player's value. Development Credit assigns all of a player's production during his evaluation window to the team that originally drafted him, regardless of whether he was later traded. In-Organization Credit only assigns credit for the specific period a player was actually on that team's roster. If a player is drafted and traded before ever playing a game, the drafting team receives none of his subsequent value under this measure, not because no development occurred while he was in their system, but because NHL production is the only signal this model can observe, and none exists yet at the point

he leaves. If a player is developed for several seasons, then traded, and continues to produce well elsewhere, the original team keeps credit only for the years it actually had him, and the new team gets credit for what happens under its own roster from that point forward. Credit can move across multiple organizations over the course of one player's career this way, and it moves specifically at the point a trade occurs, not before and not after.

This matters because giving a team permanent credit for a player that it no longer controls does not reflect what actually happened. A team's job in the draft is to identify talent. What happens to that talent afterward, whether the organization keeps developing it or hands that responsibility over to someone else through a trade, is a separate outcome, and the model needed to be able to tell those two things apart.

3.4 Player Value Metric

Rather than weigh goals, assists, plus-minus, and every other factor that plausibly matters to a player's value myself, and risk quietly building my own assumptions about what “good” looks like into the results, I chose to rely on Point Shares, an already-established, independently validated metric. Point Shares already accounts for the factors I felt were necessary, in a way that has already been tested and trusted, rather than something I would have been inventing from scratch. Using it meant I was not creating new information and hoping it would hold up. I was applying something that already existed and was reliable. No single public hockey metric perfectly captures player value; Point Shares was selected not because it eliminates measurement error, but because it provides a consistent, position-adjusted framework for comparison across more than 2,300 players.

Point Shares was also the more defensible choice for a second reason: a defenseman's contribution is not measured on the same scale as a forward's offensive production, which matters given how

differently those positions create value on the ice. Building my own formula would have meant either ignoring that distinction or trying to solve it myself, and I would rather rely on an already-validated metric than risk quietly building my own assumptions about what “good” looks like into the results.

Players are grouped into three position categories in this project, Forward, Defenseman, and Goalie, because a given draft slot means something different depending on position entirely. The odds of a third-round defenseman becoming an NHL regular are not the same as the odds for a third-round forward, and treating them identically would have distorted the expectation baseline for both.

I also considered other ways to quantify a player's success before settling on Point Shares, including postseason awards, Stanley Cup appearances, All-Star selections, All-NHL team honors, and simply time spent in the league. I ultimately did not use any of these, for a specific reason in each case. Awards and All-Star recognition are largely redundant with what Point Shares already captures, since a player having an award-worthy season already shows up as a clear outlier in his underlying production. Including awards on top of that risked counting the same performance twice under two different labels. Games played, and time spent in the league more broadly, is addressed elsewhere in this project through the confidence-weighting system rather than folded directly into the value metric itself, since simply staying in the league is a different signal than performing above expectation while there. None of these alternatives were rejected because they lack any value as information. They were rejected because they did not accurately answer the specific question this project was built around, and Point Shares, on its own, did.

3.5 Expected Pick Value

The bucket sizes, groups of 5 picks for skaters and 20 picks for goalies, came from thinking about how many players actually get drafted at each position. Far more skaters are drafted than goalies in any given year. There are twelve forward spots and six defense spots per team, compared to roughly two goalie spots per team when counting both starters and backups. If goalies were grouped into the same 5-pick buckets as skaters, most buckets would only contain zero or one goalie, which is not enough to build a reliable average. Using wider, 20-pick buckets for goalies specifically keeps each group large enough to actually be meaningful.

The smaller bucket size for skaters matters for a related reason. A player projected in the top five picks is generally expected to perform much more similarly to another top-five pick than to a player drafted picks six through ten, since draft projections shift quickly at the very top of the draft. Narrower buckets for skaters keep the expectation baseline tight where it needs to be, without losing the reliability that comes from grouping picks together rather than treating every single slot as its own isolated data point. I tested and compared different bucket sizes before settling on this combination, and this split gave the most sensible balance between precision and having enough players in each group to trust the result.

I chose to build the Expected Value curve directly from this project's own dataset rather than applying an existing curve like Tulsky's directly, because the two are actually answering different questions. Tulsky's curve estimates what a draft slot is worth in general, based on historical outcomes across the league as a whole. It is not built around individual, actual players and what specifically happened to them once they were drafted. This project needed a baseline built from real, verified outcomes for the exact players in this exact dataset, not an outside estimate applied on top of them.

Building the curve empirically also meant the expectation baseline reflects this project's own scope directly, the same 2010 through 2020 window, the same evaluation windows, the same value metric, rather than importing assumptions from a different era, a different set of draft classes, or a different underlying value metric entirely. The result is a baseline grounded in what players picked in a given range and position produced during this specific study, not a general-purpose curve applied after the fact.

3.6 The Organizational Development Index (ODI)

Every drafted player's Player Development Score gets scaled by a confidence weight before it factors into his team's total, based on how many games he actually played during his evaluation window. The idea is simple: a player who only appears in 20 NHL games and happens to score well during that stretch should not count the same as a player who has proven himself over a genuinely large sample. A short hot streak is not the same thing as real, established NHL value, and without some adjustment for sample size, a small number of lucky games could swing an entire team's ranking. I adopted a threshold of 300 games, roughly half of a full 6-year career, as the point where a player's production reflects a real trend rather than a small sample. Looking back at this decision honestly, 300 was chosen as a reasonable, intuitive rule of thumb rather than something derived statistically from where variance in a player's production stabilizes as games played increases. The sensitivity test run later in this project showed the model's core rankings hold up whether this threshold is set lower or higher, which is reassuring, but that robustness is not the same as proof the number itself is correct. A more rigorous future version would derive this threshold empirically rather than from intuition, and would also account for the fact that “half a career” represents a different number of games in the 6-year window than it does in the 8-year window, a distinction the current model does not make.

At the team level, this project reports two separate numbers, because they answer two different questions. ODI-Total adds up every player's weighted score for a given team, and answers how much total surplus value that organization has gotten from its draft picks overall. ODI-Efficiency divides that total by the number of valid picks the team actually made, excluding any forfeited selections, and answers how good the organization is, pick-for-pick, at development specifically. I consider ODI-Efficiency the headline number for this project, not ODI-Total, since it directly reflects the second half of this project's research question, how much retained value an organization gets per pick, without rewarding a team simply for having more draft picks to work with. ODI-Total is still reported, since it answers a real, related question about total value gained, but it was never the primary headline metric the way Efficiency was.

4. Results

How to Read These Numbers

Before presenting the findings, it is worth explaining plainly what the central number in this paper, ODI-Efficiency, actually means, since it is used throughout the rest of this paper without further definition. ODI-Efficiency is built around a zero point: zero means an organization's drafted players produced exactly what their draft position predicted, no more and no less. A positive number means the organization's players outperformed that expectation on average, and a negative number means they underperformed it. There is no fixed maximum; the scale is relative to what actually happened across the 2010-2020 dataset, not a percentage or a 0-to-100 index. In practice, across all 31 organizations in this study, ODI-Efficiency in the 8-year window ranges from roughly 0.3 to 4.2, so a team in the 3s or 4s is meaningfully outperforming draft-position expectations, a team near 1 is performing close to what would be expected on average, and a team below 1 is

underperforming it. The same logic applies to the positional breakdowns in Figure 3 and the late-round figure below, just calculated within a narrower subset of picks.

The final 8-year ODI-Efficiency rankings placed Anaheim Ducks first (4.2), followed by Boston Bruins (3.9), Carolina Hurricanes (3.5), Calgary Flames (3.2), and St. Louis Blues (3.2). Both organizations ranked in the top three under both the 6-year and 8-year evaluation windows independently.

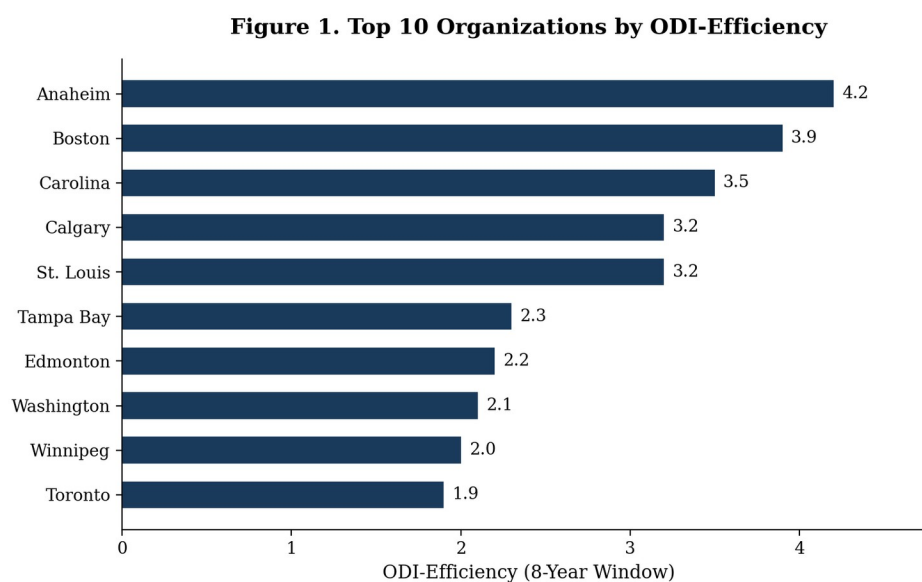


Figure 1. Top 10 organizations by ODI-Efficiency, 8-year evaluation window.

Filip Forsberg, Seth Jones, and Samuel Girard represent the largest gaps between Development Credit and In-Organization Credit in the dataset. Forsberg was drafted by Washington (11th overall, 2012) and traded before playing a game for the organization; his In-Organization value for Washington is 0.0. Jones was drafted by Nashville (4th overall, 2013), produced 10.1 Point Shares as a Predator before being traded to Columbus in 2016, and produced 52.2 total Point Shares within the evaluation window, a gap of 42.1 points. Girard shows a similar pattern following his trade

from Nashville to Colorado. Nashville's overall ranking falls from 1.9 (Development Credit) to 0.2 (In-Organization Credit), the largest shift in the dataset. Washington's ranking falls from 2.1 to 0.7.

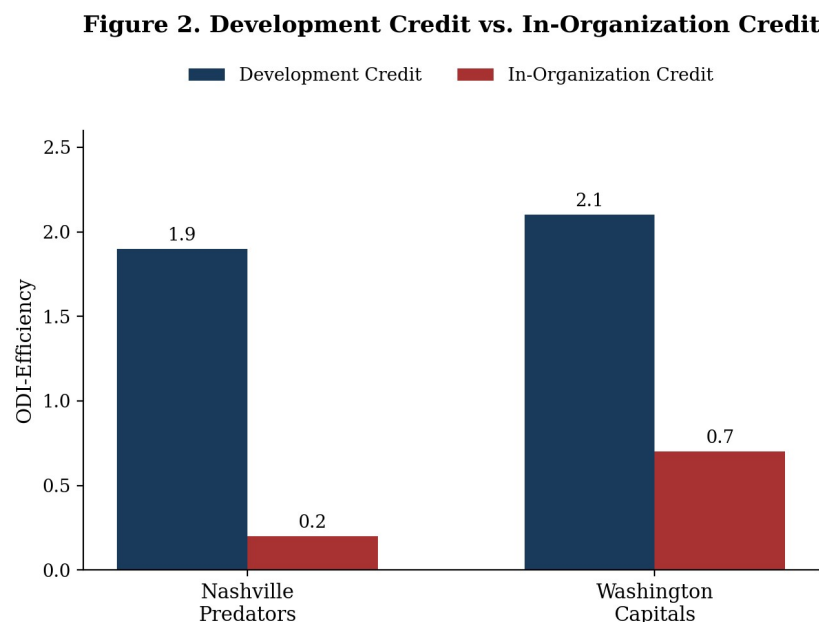


Figure 2. Development Credit vs. In-Organization Credit for the two largest gaps in the dataset.

Detroit Red Wings rank 27th of 31 organizations (0.7 efficiency) despite having the largest sample size in the study, 94 total picks. Vegas Golden Knights rank last (0.3 efficiency) with 20 picks, the smallest sample size in the study, reflecting only 3-4 completed draft classes as a 2017 expansion franchise.

Positional breakdowns show Anaheim ranking first in both defense (5.89 efficiency) and goaltending (15.92 efficiency) development, and Carolina ranking first in forward development (3.97 efficiency). Chicago's data shows a forward efficiency of 3.46 against a defense efficiency of -0.54, the largest internal positional gap of any team in the dataset.

**Figure 3. Positional Development Efficiency by Organization
(sorted by approximate overall efficiency, descending)**

	Forward	Defense	Goalie
Anaheim	1.12	5.89	15.92
Boston	3.46	4.74	3.38
Carolina	3.97	2.48	3.92
Calgary	3.26	3.62	0.20
St. Louis	3.14	4.15	0.76
Tampa Bay	3.25	0.44	3.23
Edmonton	1.76	3.12	1.48
Washington	2.98	1.18	1.13
Winnipeg	2.15	0.85	5.03
Toronto Leafs	2.38	1.51	0.06
Nashville	1.82	2.10	1.63
Los Angeles	1.41	3.29	-0.08
Dallas Stars	0.61	3.07	3.28
Chicago	3.46	-0.54	0.00
Colorado	1.38	2.30	0.61
Vancouver	0.95	2.49	1.12
New York	1.49	1.56	0.98
Ottawa	1.88	1.02	-0.07
Columbus Blue Jkts	1.36	1.63	1.34
San Jose	1.93	0.37	0.00
New Jersey	1.78	0.26	1.11
Philadelphia	0.44	1.79	3.09
Pittsburgh	1.50	-0.28	5.52
Buffalo	0.10	2.31	2.48
Minnesota	1.29	0.09	0.78
New York	-0.74	2.37	0.40
Detroit Wings	0.21	0.88	2.71
Florida	0.45	0.51	0.54
Montreal	0.02	1.13	0.20
Arizona	-0.09	0.98	0.85
Vegas Golden Kts	-0.05	1.22	0.00

Figure 3. Positional development efficiency for all 31 organizations, by Forward, Defense, and Goalie. Organizations are sorted by an approximate overall efficiency (total weighted value divided by total picks across all three positions), descending.

In the late-round drafting analysis (rounds 4-7, estimated from pick number), Ottawa Senators rank first (2.31 efficiency), followed by Carolina Hurricanes (2.26) and Calgary Flames (2.11). Pittsburgh Penguins rank last (0.18 efficiency) with 61 total picks used, the second-lowest pick count in the study.

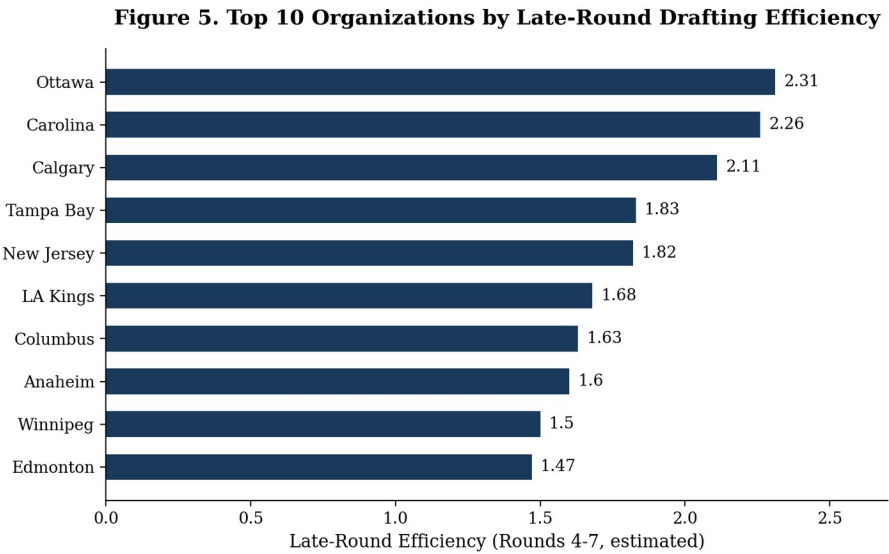


Figure 5. Top 10 organizations by late-round (rounds 4-7) drafting efficiency.

The sensitivity test, re-running the full model under four alternative parameter sets, found that 9 of the 10 baseline top-10 organizations remained in the top 10 under every variant tested. Winnipeg Jets, the 10th-ranked organization in the baseline model, shifted to approximately 13th under most alternative parameter settings.

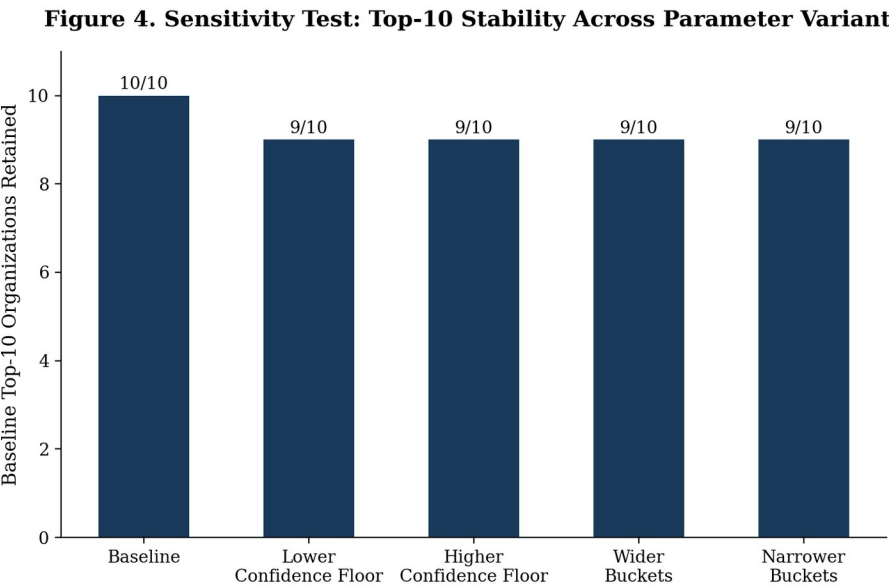


Figure 4. Number of baseline Top-10 organizations retained under each sensitivity test variant.

5. Discussion

The biggest thing this project revealed to me is that success and development are not the same thing, and they get conflated constantly. Every team has its own process, and that process is not always centered on developing the players it drafts. Some organizations are better at trading for proven talent than they are at developing prospects, and that can be a completely legitimate strategy. A team being successful does not mean the players driving that success became great specifically because of that organization's development system. Pittsburgh is the clearest example of this in the data. It was a genuinely successful, Cup-winning organization during this window, but one of the weaker performers in actual development efficiency (1.1 overall, and lowest in the entire league specifically in late-round drafting at 0.18), consistent with a team that built its success by trading for established players rather than developing its own picks. That distinction, between being good at winning and being good at developing, is easy to overlook when a team is simply judged by whether it is winning games.

Anaheim's #1 ranking is not explained by a single standout player or position. The organization ranks first in both defense and goaltending development specifically, a pattern that is harder to attribute to luck or a single outlier than a ranking driven by one exceptional forward would be. Boston and Carolina's stability across both the 6-year and 8-year windows, both ranking in the top three under each independently, similarly points toward consistent organizational strength across multiple positions rather than a result dependent on a small number of players. Detroit's decline is best explained by a combination of factors rather than a single cause, and the data available in this study cannot fully separate them. Following the Datsyuk and Zetterberg era, the organization's results declined substantially, consistent with both a deliberate rebuild and a prolonged period of underperformance following the departure of that core group. What the data does establish clearly

is that this decline is a real, sustained signal rather than a small-sample artifact, since it is built on the largest pick total of any team in this study. The one notable result from the sensitivity test, Winnipeg's shift from 10th to roughly 13th under alternative parameters, is not a meaningful finding on its own. A three-spot shift within a group of 31 organizations is a small, unremarkable movement, and the fact that it happened to cross the top-10 line says more about Winnipeg sitting exactly at that boundary in the baseline model than about any real instability in the underlying framework. The far more significant result is that the other 9 baseline top-10 organizations did not move at all across every tested variant.

This is not just an interesting way to look at an organization's own history. It has real, practical value for a front office. The positional breakdowns in particular function as a genuine diagnostic tool, not a general observation. Chicago's data shows real strength developing forwards alongside real weakness developing defensemen specifically, a forward efficiency of 3.46 against a defense efficiency of -0.54, the worst defensive development mark of any team in the study. That is not a vague sense that something needs improvement, it points toward a specific, investigable question, whether that gap comes from scouting philosophy, development staffing, or something else entirely. A team using a different strategy successfully, the way Pittsburgh did, is not automatically doing something wrong. But even a team finding success elsewhere would still generally want to be strong across the board, not just in the area currently working, because a real weakness does not stop being a weakness just because it is being masked by success somewhere else right now. A strategy built around trading for proven talent depends on having assets available to trade and cap room to support it, conditions that can change. This project does not tell every organization what to do. It shows every organization, regardless of its current success, exactly where it stands, so that gap can be an informed decision rather than a hidden one.

More broadly, this project reflects something I believe is true about the sport as a whole. There are aspects of hockey that people do not immediately think to quantify, but that does not mean they cannot be. Every metric has real limitations, and no number replaces the actual complexity of how an organization operates or the different kinds of value different approaches provide. But there is real room to go deeper analytically than most existing tools currently do, and if the data to answer a specific question does not already exist in an accessible form, it can often be built. That is genuinely what this project set out to demonstrate, not that hockey should be reduced to a spreadsheet, but that a sport people have followed their whole lives can still be understood in a new way, specific enough to help an organization identify concrete strengths and weaknesses worth investigating further.

6. Limitations

The biggest limitation of this project, in my own view, is the number of outside factors a player's career can be shaped by that this model cannot see or account for: injuries, coaching relationships, personal circumstances, timing, and other human factors that do not show up in a box score. This model measures what a player produced on the ice within a fixed window, not the full context behind why he did or did not reach his potential. No dataset like this one captures that, and it is worth stating plainly rather than treating it as a footnote.

This does not make the model less trustworthy. It asks a narrower, more specific question than “what explains everything about a player's career,” and it answers that narrower question honestly: what can be measured cleanly from what happened on the ice, within a defined and consistent standard. Context still matters and should be layered on top of what these numbers show, not treated as something the numbers already account for.

This project also has a set of clear scope limitations, some of which matter more than others.

The most significant of these is that the model only tracks NHL-level games. Every other level of competition, junior hockey, the AHL, international leagues, is completely invisible to it. This creates a real, specific blind spot that goes beyond simply saying “AHL performance isn't counted.” Consider a player drafted by one organization who spends two years developing in the AHL, gets a single season of real NHL production with that same team, is then traded, and spends the remainder of his evaluation window back in the AHL with his new organization, never returning to the NHL. Under the current model, both of his AHL stretches are entirely invisible, no value, no penalty, no signal of any kind, for either organization. Only that one NHL season counts, credited to the team he was with at the time. A player who spends the overwhelming majority of his career in the minors effectively disappears from this model except for whatever small sliver of NHL production he manages to produce. The model cannot currently distinguish between an organization that correctly judged a player was not NHL-ready and continued developing him appropriately in the minors, and an organization that mismanaged or unfairly buried a player who may have deserved a real opportunity. Both scenarios look identical under the current framework, no NHL production, and therefore no signal either way. This is a genuine, significant limitation, not a minor omission, and it is the one limitation from this project I would prioritize addressing first in any future version.

A second, related scope limitation is that undrafted free agents are excluded entirely. This is a direct consequence of how the model is built, every player's score depends on comparing his actual production to an Expected Value baseline tied to his specific draft position, and an undrafted player has no draft position for that baseline to attach to. This means an organization's success developing and signing undrafted talent, a real and increasingly relevant development pathway in today's

NHL, receives no credit at all under this framework. This is a deliberate boundary of what the model is built to measure, not an oversight, but it is still worth naming directly rather than leaving implicit.

The third scope limitation is that this project only covers draft classes from 2010 through 2020. This range was chosen as a practical starting point, broad enough to provide a meaningful sample size while remaining recent enough to reflect how organizations currently operate. Unlike the first two limitations, this one is not a structural weakness of the model itself. The year range could be expanded or shifted in a future version simply by scraping and incorporating additional draft classes using the same methodology already established here. It represents the base this project was built on, not a boundary the model is fundamentally incapable of moving past.

A related, deeper limitation is that this model still cannot fully separate scouting skill from development skill for a player who is never traded. The Development Credit versus In-Organization Credit distinction resolves this whenever a player changes teams, as shown clearly in the Forsberg and Jones cases. For a player who stays with one organization his entire career, however, the model cannot tell whether his success reflects the organization identifying a player who was always going to succeed, or real development work that improved him. This is not a gap unique to this project; it is a limit of this entire category of research, since it would require observing the same player developing under two different organizations at once. A future version could partially address this by incorporating independent pre-draft prospect rankings as a second baseline for comparison, an idea addressed further in the Future Research section, though such rankings are themselves projections rather than confirmed outcomes and would introduce their own uncertainty.

Two draft picks in this dataset were league-forfeited penalties, New Jersey's 2011 third-round pick, forfeited as part of the Ilya Kovalchuk cap-circumvention penalty, and Arizona's 2020 second-round pick, forfeited for a Combine Testing Policy violation. Both were excluded from all team totals, since neither was ever a real selection made by any organization. Two franchises relocated during this study's window, Atlanta Thrashers to Winnipeg Jets in 2011, and Phoenix Coyotes to Arizona Coyotes in 2014, and both are treated as the same continuous organization across the move, consistent with how the league itself treats relocated franchises.

The metric underlying this entire project, Point Shares, carries its own known limitation worth stating plainly: it underweights defensive value relative to offensive production, a limitation shared across most box-score-derived hockey statistics, not unique to this project. This specifically affects how much confidence should be placed in this project's defense-specific findings, including Anaheim's position as the top defensive developer in the dataset. Two concrete paths exist to address this in a future version, incorporating Hockey-Reference's separate Defensive Point Shares figure directly, or bringing in more sophisticated tracking-based defensive metrics, both discussed further in Future Research. Awards, playoff performance, and contract value were also deliberately excluded from the core formula. Awards and All-Star recognition are largely redundant with what Point Shares already captures, since an award-worthy season already appears as a clear outlier in the underlying production. Contract value reflects a different question entirely, market timing, negotiation, and salary cap conditions, closer to asset management than player development, and including it would have blended this project's actual question with a different one.

Finally, this project has real limitations tied to sample size and model stability. The Vegas Golden Knights, as a 2017 expansion franchise, have only three to four completed draft classes within this study's window, a sample too small to reliably distinguish genuine organizational weakness from

ordinary statistical noise. Detroit's similarly low ranking carries far more weight by comparison, since it is built on the largest sample size of any team in the dataset, 94 total picks, making it a real signal rather than an artifact of limited data. The sensitivity test run as part of this project's validation process found that 9 of the 10 baseline top-10 organizations remained stable across every parameter variation tested, while the 10th spot shifted modestly under some conditions, a normal and expected result rather than evidence the model is unreliable. Lastly, the draft round field used in the late-round analysis was not directly available in the scraped data, since Hockey-Reference does not store round as a per-player attribute, and was instead estimated from a player's overall pick number and the number of teams in the league that year. This estimate is accurate for the large majority of picks, but may be off by one selection near a round boundary in years affected by compensatory picks.

7. Future Research

This project represents a first version, not a finished, permanent answer, and there are specific, identified next steps worth pursuing if this framework continues.

One of the most significant additions would be incorporating AHL and other non-NHL performance into the model, likely through an NHL Equivalency-style conversion, so a player's time in the minors is no longer invisible to the framework regardless of which organization is responsible for it. Closely related, and more immediately achievable, would be improving how the model handles defensive value specifically, either by incorporating Hockey-Reference's separate Defensive Point Shares figure directly, or by bringing in more sophisticated tracking-based defensive metrics from a source built specifically for that purpose. The confidence-weighting threshold used throughout this project, currently a reasonable but intuitively chosen 300 games,

should be derived empirically in a future version rather than assumed, and should account for the fact that a full career looks different in length across the 6-year and 8-year windows. The sensitivity testing already performed in this project shows the core rankings are stable, and a future version could build on that foundation with additional statistical tools, confidence intervals around team scores, formal significance testing to characterize gaps like the one between Anaheim and Boston more precisely, and resampling techniques to test ranking stability from a different angle than the parameter-based approach used here. Finally, incorporating independent pre-draft prospect rankings as a second comparison baseline would allow a future version to begin separating scouting skill from development skill even for players who are never traded, addressing the deepest and most difficult limitation this project identified.

Zooming out from ODI specifically, this project points toward something broader about where sports analytics can still go. The framework built here, an empirically derived expectation baseline compared against actual outcomes, adjusted for the specific circumstances a player or organization operates under, is not inherently limited to NHL draft classes from 2010 through 2020. The same underlying approach could, in principle, extend to a larger span of draft history, or eventually to organizational development questions in other sports entirely, provided the same care is taken to define the question precisely and remain honest about what the resulting model can and cannot capture. This project is not a claim that every part of a sport can or should be reduced to a number. It is a demonstration that a specific, well-defined question about a sport, one that has mostly lived as opinion and reputation, can be answered with real evidence instead, and that when that evidence gets built carefully and questioned honestly, it can genuinely change how an organization understands itself.

8. Conclusion

This project set out to answer a specific question: how efficiently does an NHL organization convert the players it drafts into NHL value relative to what their draft position predicts, and how much of that value does the organization actually retain versus lose through trades? The results indicate that this can be measured, and that doing so changes real conclusions, as shown directly by the Nashville and Washington cases, where a large share of apparent drafting success did not correspond to actual in-organization development.

The Organizational Development Index demonstrates that this separation is not merely theoretical. It produces stable, cross-validated rankings, holding consistent across two independent evaluation windows and a battery of sensitivity tests, and it produces specific, actionable findings, including clear positional strengths and weaknesses within individual organizations that a front office could investigate directly. The project's own internal audit, identifying and correcting a scraping error that had zeroed out real production for every goalie in the dataset, further demonstrates that the resulting model reflects verified data rather than an unexamined first output.

Building this project required developing the technical workflow necessary to collect, validate, and analyze a dataset of over 2,300 players, including auditing the model's own results rather than accepting them at face value, and revising methodological assumptions when they did not hold up to scrutiny. That process is part of what makes this project's conclusions defensible: it does not replace the research findings themselves, it is the reason those findings can be trusted. The core result stands on its own: a sport largely evaluated through reputation and anecdote can be evaluated with real, transparent evidence instead, and doing so changes the picture in ways worth taking seriously.

Appendices

Appendix A: Core Formula Reference

Player Development Score (PDS) = Actual Value (windowed Point Shares) – Expected Value (for pick bucket and position)

This is the core comparison at the heart of the model: how much Point Shares value a player actually produced within his evaluation window, minus how much a player picked at that position and draft slot was expected to produce on average. A positive PDS means the player outperformed expectation; a negative PDS means he underperformed it.

Confidence Weight = $\min(\text{Games Played in window} / 300, 1.0)$

This scales a player's score down if he has not played many games, so a small, lucky sample cannot swing a team's ranking. A player with 300 or more games gets full weight (1.0); a player with 150 games gets half weight (0.5); a player with very few games gets very little weight.

Weighted PDS = $\text{PDS} \times \text{Confidence Weight}$

This is a player's final, individual contribution to his team's score, his raw performance above or below expectation, discounted by how much NHL evidence actually exists for that performance.

ODI-Total = $\Sigma \text{Weighted PDS across all of a team's picks}$

This adds up the Weighted PDS of every player a team drafted, giving the organization's total surplus value across all of its picks combined.

ODI-Efficiency = $\text{ODI-Total} \div \text{number of picks used}$

This divides ODI-Total by how many valid picks the team made, turning a raw total into a per-pick rate. This is the headline metric used throughout this paper, since it allows fair comparison between teams with very different numbers of draft picks. See “How to Read These Numbers” at the start of the Results section for how to interpret the resulting scale.

Appendix B: Position Group and Bucket Definitions

Position groups: Goalie (if “G” appears in listed position), Defenseman (if “D” appears and not already classified as goalie), Forward (all remaining positions)

Skater pick buckets: groups of 5 consecutive picks (1-5, 6-10, 11-15, etc.)

Goalie pick buckets: groups of 20 consecutive picks, due to the substantially smaller number of goalies drafted relative to skaters

Appendix C: Data Cleaning Log

One forfeited pick identified and removed from all totals: New Jersey Devils, 2011 third round (pick 69), forfeited as part of the Kovalchuk cap-circumvention penalty.

One additional forfeited pick confirmed present but requiring no removal, since Hockey-Reference omitted the row entirely rather than inserting a placeholder: Arizona Coyotes, 2020 second round (pick 49), forfeited for a Combine Testing Policy violation.

Franchise name mapping applied for continuity across relocation: Atlanta Thrashers merged with Winnipeg Jets (2011 relocation); Phoenix Coyotes merged with Arizona Coyotes (2014 rebrand).

Thirteen player records identified with incomplete data due to a rate-limiting interruption during collection; all thirteen were re-scraped and verified complete before final analysis.

A scraping error affecting all goalie records, caused by goalies using a separate Hockey-Reference statistical table (goalie_stats) with different column names (gps for Point Shares, goalie_games for games played) than skaters, was identified, corrected, and all affected records were re-scraped and re-verified.

Appendix D: Code and Data Availability

The full Python scraping and analysis pipeline, along with the complete consolidated results workbook (ODI_Master_Results.xlsx), are available in the accompanying project repository.

Annotated Bibliography

Kubatko, J. “Calculating Point Shares.” Hockey-Reference.com. https://www.hockey-reference.com/about/point_shares.html

The original methodology page defining Point Shares, the core player value metric used throughout this project. Referenced to understand exactly what the metric does and does not capture, including its known limitations around defensive value.

Tulsky, E. “NHL Draft: What does it cost to trade up?” Broad Street Hockey, April 25, 2013. <https://www.broadstreethockey.com/post/nhl-draft-pick-value-trading-up/>

The original article establishing a draft pick trade-value curve based on historical draft-day trades. Referenced during the design of this project's evaluation windows and as a point of contrast in the Literature Review, since Tulsky's curve estimates pick value from trade behavior rather than from actual player outcomes.

“What Does It Mean To Draft Perfectly? An Evaluation of Draft Strategy.” Joseph Wharton Research Scholars, University of Pennsylvania.

An academic paper that independently used Hockey-Reference's Point Shares metric to evaluate NHL draft outcomes, providing real precedent for this project's choice of value metric and general approach.

“NHL Modifies Ilya Kovalchuk Penalty Against Devils.” NHL.com.

Official league statement confirming the New Jersey Devils' 2011 third-round pick forfeiture as part of the salary cap circumvention penalty tied to the Ilya Kovalchuk contract. Used to verify and correctly classify a forfeited pick identified during data cleaning.

“NHL Announces Sanctions to Coyotes for Violating Combine Testing Policy.” NHL.com.

Official league statement confirming the Arizona Coyotes' 2020 second-round pick forfeiture for a Combine Testing Policy violation. Used to verify a second forfeited pick identified while auditing per-year draft totals against known historical figures.

Hockey-Reference.com.

The primary data source for this entire project, providing NHL Entry Draft records and individual player statistics for all 2010-2020 draft classes.

hockeydb.com.

Used to independently verify total draft pick counts per year during data validation, confirming the accuracy of the scraper's collection against a second, independent source.